

LRU Cache Test Cases (4-way, 4-set)

Set 0 - Filling

Step	Tag	Index	Expected Result
1	0x001	00	Miss → Way0
2	0x002	00	Miss → Way1
3	0x003	00	Miss → Way2
4	0x004	00	Miss → Way3

Set 0 - LRU Update

Step	Access	Expected
5	Tag 0x002	Hit → becomes MRU

Set 0 - Eviction

Step	Tag	Expected
6	0x005	Miss → Evict LRU (0x001)

Set 0 - Validation

Step	Access	Expected
7	0x001	Miss (evicted)
8	0x002	Hit

Set 2 - Filling

Step	Tag	Index	Expected
9	0x011	10	Miss
10	0x012	10	Miss
11	0x013	10	Miss
12	0x014	10	Miss

Set 2 - LRU Update

Step	Access	Expected
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13	0x012	Hit
14	0x013	Hit

Set 2 - Eviction

Step	Access	Expected
15	0x015	Miss → Evict 0x011

Set 2 - Validation

Step	Access	Expected
16	0x011	Miss
17	0x012	Hit
18	0x013	Hit